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ORIGINAL RESEARCH

Multi-task few-shot learning with composed data augmentation for image classification

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Abstract

Few-shot learning (FSL) attempts to learn and optimise the model from a few examples on image classification, which is still threatened by data scarcity. To generate more data as supplements, data augmentation is considered as a powerful and popular technique to enhance the robustness of few-shot models. However, there are still some weaknesses in applying augmentation methods. For example, all augmented samples have similar semantic information with respect to different augmented transformations, which makes these traditional augmentation methods incapable of learning the property being varied. To address this challenge, we introduce multi-task learning to learn a primary few-shot classification task and an auxiliary self-supervised task, simultaneously. The self-supervised task can learn transformation property as auxiliary self-supervision signals to improve the performance of the primary few-shot classification task. Additionally, we propose a simple, flexible, and effective mechanism for decision fusion to further improve the reliability of the classifier, named *model-agnostic ensemble inference* (MAEI). Specifically, the MAEI mechanism can eliminate the influence of outliers for FSL using non-maximum suppression. Extensive experiment results demonstrate that our method can outperform other state-of-the-art methods by large margins.

KEYWORDS

data augmentation, few-shot learning, multi-task learning, non-maximum suppression, self-supervised learning

1 | INTRODUCTION

In recent years, deep neural networks have already outperformed humans on image classification with enormous labelled samples supported [1, 2], which may be against humans learning behaviour. Humans, however, possess a fast adaptive capacity of recognising novel classes with a handful of annotated samples [3]. For example, a child can easily generalise the concept of cats and quickly recognise them in reality with only one picture from a book or the Internet [4]. In contrast, existing data-driven deep learning algorithms lag far behind humans in versatility and generalisation ability. Therefore, how to construct human-like algorithms and perform visual recognition tasks under data scarcity has important practical value, which also has attracted extensive research interests.

As a technology driving the human-like algorithm boom, few-shot learning (FSL) [4] attempts to challenge the image classification task with a hand of labelled samples [5, 6]. The main paradigm of FSL is training a model on the base classes and requiring it to accurately classify the novel classes with a limited number of examples, which is still threatened by data scarcity. Data augmentation is considered as a powerful and popular technique to obtain enough annotated samples, which is widely used in the image classification task to enhance the robustness of models. However, all augmented samples have similar semantic information (in Figure 1a) with respect to different augmented transformations, which makes these traditional augmentation methods incapable of learning the property being varied.

Self-supervised label augmentation (SLA) [7] uses a double-label technique to tackle this problem, which adds auxiliary self-

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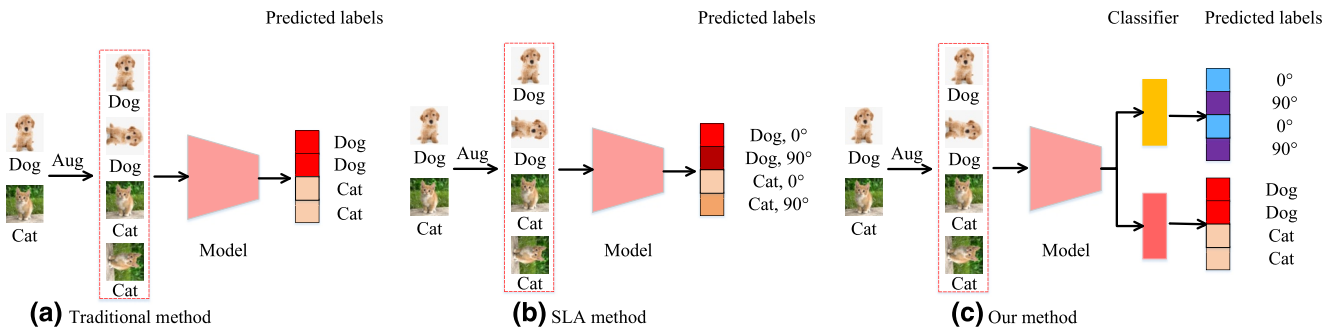

FIGURE 1 An overview of our multi-task learning and previous approaches.

supervised labels to represent the transformation differences. As shown in Figure 1b, all augmented images have two different types of labels, the same category labels and the unique self-supervised labels. Therefore, SLA can avoid the risk from data augmentation by learning a more fine-grained classification task than the primary classification task. However, the problem plagued by the double-label technique is the increasing complexity of primary classification tasks, and this complexity could be magnified as the classified category increases.

In this paper, we introduce a multi-task framework [8] to tackle aforementioned issues, which does not increase the complexity of the primary classification task. As shown in Figure 1c, our proposed method consists of a primary classification task and a self-supervised task [9]. The self-supervised task can learn the transformation property as auxiliary self-supervision signals to improve the performance of the primary few-shot classification task. The complexity of each classification task is not increasing for our multi-task framework, since the self-supervised task is parallel to the primary classification task. Instead of using a single data augmentation operation [8, 9], we attempt to predict three data augmentation operations as self-supervision signals and explore the best combination augmentation operation for few-shot image classification with self-supervised tasks. Specifically, we find that using rotation [10] and mixup [11] operations with self-supervision signals can achieve the best performance than other combinations.

In order to further improve the reliability of the classifier, we propose a *model-agnostic ensemble inference* (MAEI) mechanism for decision fusion, which can adapt to any metric-based FSL method without fine-tuning. The MAEI mechanism aims to produce a reliable ensemble result using the non-maximum suppression (NMS) strategy [12]. Specifically, our MAEI mechanism suppresses non-maximum predictive scores and integrates the maximum one as the final output. Extensive experiments demonstrate that our method outperforms other existing state-of-the-art FSL approaches.

Our main contributions can be summarised as follows:

- We introduce a multi-task learning framework for FSL, which consists of a primary classification task and an auxiliary self-supervised task. The auxiliary self-supervised task makes models learn transformation property from these augmented samples. Therefore, our multi-task learning framework can eliminate the invariant property of the augmented images with respect to the transformations. And our method can avoid the increasing complexity caused by the double-label technique since the self-supervised task is parallel to the primary classification task.
- We propose a composed augmentation strategy with a self-supervision signal to improve the performance of FSL. We attempt to predict three data augmentation operations as self-supervision signals, and find the best combination augmentation operation for few-shot image classification.
- We propose a MAEI mechanism for decision fusion to further improve the reliability of the classifier, which is a simple, flexible, and effective mechanism for FSL in the image classification. The main idea of our MAEI mechanism is suppressing the non-maximum similarity scores and only preserving the maximum one. And this mechanism can adapt to any metric-based FSL methods without fine-tuning.
- We conduct extensive experiments on different benchmark data sets. Experiment results demonstrate that our method outperforms other state-of-the-art FSL methods. Additionally, we perform substantial ablation experiments to further validate the importance of our self-supervised task and MAEI mechanism.

2 | RELATED WORK

Few-Shot Learning: With the emergence of FSL, various approaches have been proposed to tackle the issue of learning with few samples [13–15]. In general, these methods can be divided into two main directions: transfer-learning methods [16, 17] and meta-learning methods [18].

Early works in FSL utilise feature transfer [17] to bridge the gap caused by limited data. A standard transfer based approach pre-trained the embedding function on base classes first, then fine-tuned the parameters of the embedding function for novel classes. The limitation of transfer-learning approaches is that they seriously rely on the relativity between the base classes and the novel classes. The pre-trained models may fail to fine-tune with novel tasks, so these methods have the risk of over-fitting when the new classes have low similarity with the base classes.

Meta-learning [19] attempts to learn an initial base learner on various subtasks that are able to fast adapt to novel classes. These methods can be roughly divided into gradient-based approaches

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[20, 21] and metric-based approaches [10, 22, 23]. Gradient-based approaches can adapt to novel classes without pre-training, but still need fine-tuning. To efficiently generalise to novel classes, metric-based approaches perform classification tasks using a feature extractor as the classifier, which attempts to map the embedding of same category to a shared metric space [24]. For example, Matching network [22] uses cosine distance to produce a weighted probability distribution via a softmax function. Whereas, Prototypical network [23] measures the unlabelled images with the corresponding class prototype using Euclidean distance. RelationNet [10] obtains relative representations via feature connection operation and transforms them into similarity scores by a non-linear CNN classifier. Despite the considerable strides make by the above works, the predicted labels are intrinsical uncertainty due to lacking samples. In this paper, we try to shed some light on data augmentation with multi-task learning, which can make the embedding function to learn continuous semantic information from augmented images.

Data Augmentation: Data augmentation is a classic technology during the data preprocessing stage, which can enhance the robustness of the model by producing more availably augmented samples and corresponding fake labels from the original data set [17]. Several approaches try to modify the appearance and background of original images by duplicating them into some other samples [4]. For example, Delta-encoder [25] uses various auto-encoders to transfer variation between the original images and some similar samples. FATTEN [26] further proposes to learn a continuous attribute subspace that can be used to bring in any attribute variations to the original images. However, these approaches are data-driven methods, which are costly and overwhelmingly rely on the prior knowledge of FSL tasks.

Handcrafted rule [4] is another effective technique for data augmentation, which can be divided into geometric transformation and adding noise. The classical geometric transformation approaches often obtain more complex augmented images by rotating, cropping, flipping, cutting out and so on [27, 28]. However, these methods cannot produce auxiliary information because these operations can not change the relative position between pixels. Therefore, the noisy-based approaches attempt to produce more available samples with auxiliary pixel information such as colour permutation [7] and mixup [11]. In this paper, we combine the advantage of these handcrafted rules and explore a best composed augmentation strategy with self-supervision singles for the FSL task.

Self-Supervised Learning: Self-supervised learning [9, 29–31] performs pretext tasks and attempts to enhance the capability of representations without auxiliary annotated labels [7], which is widely used in semi-supervised learning [32], unsupervised learning [33], and generative adversarial networks [8, 34]. For example, the clustering-based methods [35–37] propagate predicted labels by the cluster indices to obtain higher confidence samples. Alternatively, transformation-based methods [8, 9] use the self-supervised signal to assign the transformation of inputs. Our self-supervised module belongs to the transformation-based category, which predicts the transformation property to help the classification task.

3 | METHODS

In this section, we will show our method in detail. Firstly, we introduce the normal definition and the detail of our training strategy. Secondly, we describe the overall architecture of our framework, including a data augmentation module and a multi-task loss module. Finally, we illustrate the ensemble inference mechanism, which can further improve the reliability of our classifier.

3.1 | Problem statement

FSL attempts to accurately classify novel categories with few training examples similar to the weakly semi-supervised task. Traditionally, the classification models learn from a data set D, which consists of three sub-data sets: $\{D_{\text{train}}, D_{\text{val}}, D_{\text{test}}\}$. Due to the limited supervised information, traditional deep learning models are difficult to bridge the gap between the embedding space of the base (training) classes and the novel (test/val) classes. Following the commonly used strategy by some recent works [10, 23], we alleviate this problem using a meta-learning method. Concretely, for each training episode, we first sample N different classes randomly from the D_{train}, then we select K images and B images from each class to construct the support set $S = \{(x_{i,j}, y_{i,j})\}, i \in (1, N), j \in (1, K)$ and the query set $Q = \left\{\left(\tilde{x}_{i,j}, \tilde{y}_{i,j}\right)\right\}, i \in (1, N), j \in (1, B)$ respectively. We define the above settings as a *N-way K-shot* task. Generally, we define the $N = 5$ and $K = 1$ as a 5-*way* 1-*shot* task and the $N = 5$ and $K = 5$ as a 5-*way* 5-*shot* task. And we verify the performance of our method by performing these two tasks. The detailed settings of the query set can be seen in the Section 4.1.

As shown in Figure 2, we attempt to jointly train the primary classification task and the auxiliary self-supervised task in our multi-task framework. Firstly, we perform data augmentation in the support set and obtain the augmented set $S_{\text{aug}} = \{(x_{i,j}, y_{i,j})\}, i \in (1, N), j \in (1, K * M)$, where M is the number of augmented samples for per original sample, and the $M = 4$ in Figure 2. Specifically, we utilise the composed data augmentation with a self-supervision signal to enhance the complexity of the input data. Secondly, the embedding function f_ϕ aims to map the input x to the embedding feature z, $z = f_\phi(x) \in \mathbb{R}^{C \times W \times H}$. Different from Prototypical network [23], we use a couple of shared parameters ResNet-12 [5, 38] as the embedding function to extract the embedding features of the primary classification task and the self-supervised task respectively.

As shown in Figure 3, the embedding function consists of four ResNet blocks, and each ResNet block has three Conv2d layers, three BN layers, three ReLU layers, and a Max pooling layer. Then, the prototypical module could perform a similarity metric between the support features and the query feature using Euclidean distance. During the training stage, we aggregate all the scores to only one score by average weighting. And during the testing stage, we design a MAEI mechanism to improve the classification accuracy, which has a NMS inference. The main idea of NMS is suppressing the non-maximum

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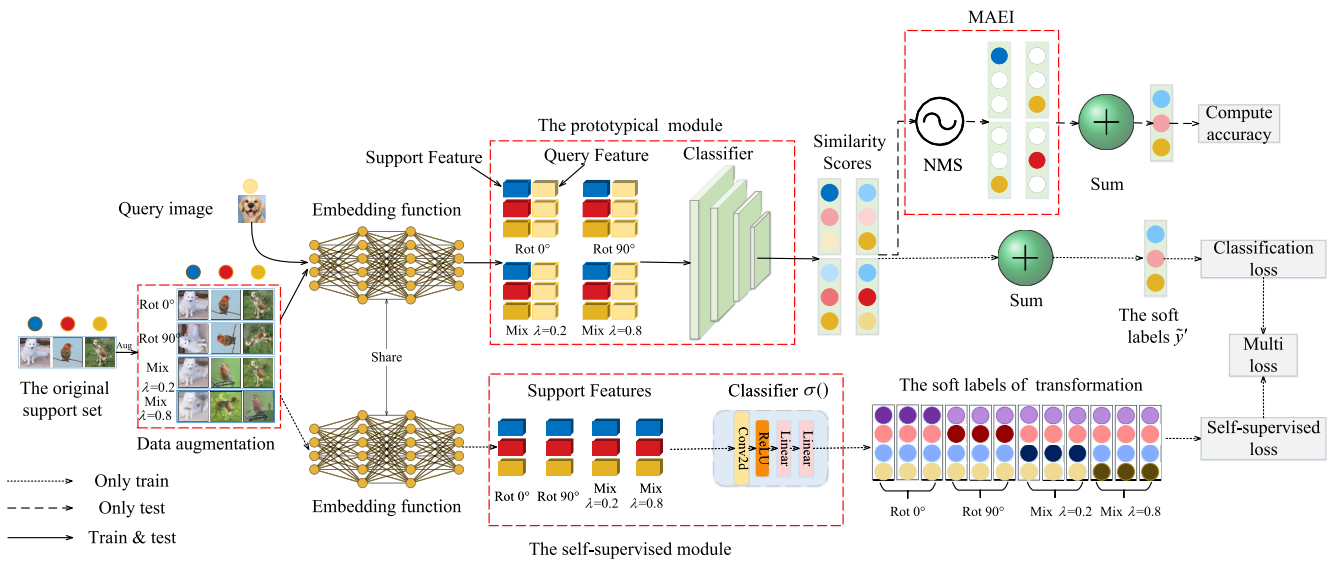

FIGURE 2 The schematic diagram of our framework in a 3-*way* 1-*shot* task with a single query sample.

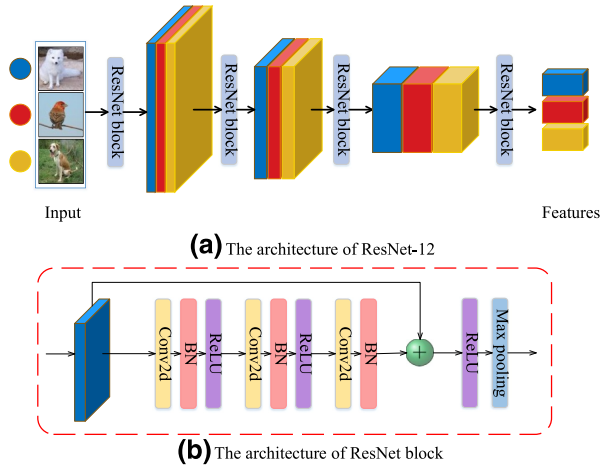

FIGURE 3 The detail architecture of the embedding function ResNet-12. (a) The architecture of ResNet-12. (b) The architecture of one ResNet block.

similarity scores. For the self-supervised module, we define a linear classifier $\sigma : \mathbb{R}^{C*H*W} \rightarrow \mathbb{R}^{M}$ to predict the transformations applied to the support set. Finally, we jointly train the primary classification task and the self-supervised task by combining their loss with a hyper-parameter β, which will be discussed in the Section 4.4.

3.2 | Composed data augmentation

Data augmentation is usually performed in the data preprocessing stage. In this subsection, we attempt to predict three data augmentation operations as self-supervision signals, and explore the best combination augmentation operation for few-shot image classification. Specifically, we will discuss a combination from rotation [7, 10], mixup [11], and colour permutation [7] for the FSL tasks. Although these three data augmentation operations are widely used in the image classification, we want to further improve the performance of the model by using multiple data augmentations with self-supervision signals.

In general, these three single data augmentation operations have different merits. For example, colour permutation and mixup can produce auxiliary noise information because they change either the value of pixel or the relative location between pixels. And the rotation operation can enlarge the original image by changing the geometric architecture. Therefore, we propose a composed data augmentation strategy to further utilise these merits.

To explore the best composed data augmentation with self-supervision signals, we select different data augmentation operations to build a composed data augmentation operation. Each composed data augmentation operation can be formalised as follows:

$$S_{\text{aug}} = \left\{ x' | x' = H\left(x_{i,j}\right) \right\}_{i,j=0}^{i=N, j=K} \tag{1}$$

where $H()$ is a composed augmentation operation, consisting of two or three single data augmentation operations, and x' is a subset of S_{aug}.

We define these single data augmentation operations as sub-policies, and all sub-policies can compose four different $H()$. For each $H()$, we randomly select two augmented images from each sub-policy, therefore, x' contains M samples, instead of one, and the M is a variable for different combinations, as shown in Table 1. For example, the value of $M = 4$ for $H()$ with two sub-policies and $M = 6$ for $H()$ with three sub-policies.

We finally find the $H()$ with mixup and rotation is the best combination operation with self-supervision signals by performing numerous experiments, which will be discussed in the Table 4. In order to better describe our method in the rest

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TABLE 1 Different composed augmentation operations with different sub-policies and the corresponding value of M

Col	Rot	Mix	M
√	√		4
√		√	4
	√	√	4
√	√	√	6

subsections, we divide the S_{aug} into M groups, and define S_m as the mth group support images.

3.3 | Multi-task for few-shot learning

For each support image, we transform it with the composed data augmentation operation to enhance the complexity of augmented images. However, all augmented samples have similar semantic information with respect to different augmented transformations, which makes these traditional augmentation methods incapable of learning the property being varied. To tackle this problem, we introduce a multi-task learning framework to learn a primary classification task and an auxiliary self-supervised task. Specifically, in the primary classification task, our model utilises a prototypical classifier module to discriminate the query set based on distance metrics. In the self-supervised task, we attempt to make our model remember the transformations by SLA.

Our multi-task loss consists of two parts produced by the prototypical classifier module [23] and the self-supervised classifier module. The prototypical classifier module is a pioneering architecture based on the metric learning approach for FSL [23]. Due to the simplicity and great generalisation ability, it has become a basic algorithm and generalised various variants [7]. Several variants similar to Prototypical network [23], we attempt to calculate the prototypes P_m by averaging the support features for the mth group, as in Equation (2).

$$P_{m,n} = \frac{1}{|S_{m,n}|} \sum_{x_{n,j} \in S_{m,n}} f_\phi(x_{n,j}) \tag{2}$$

where $S_{m,n}$ denotes the set of support samples within the class n in the mth group, and $n \in \{1, 2, \ldots, N\}$. $P_{m,n}$ denotes the prototype of the class n in the mth group. And f_ϕ denotes the embedding function with learned parameter ϕ, which generates the support embedding features $f_\phi(x_{n,j}) \in R^{C*H*W}$, $x_{n,j} \in S_{m,n}$ for *N-way* tasks. Additionally, the embedding function could produce $M * N$ prototypes, because we augment the original support set S to S_{aug} during the data preprocessing.

Then, we utilise the Euclidean distance metric $d(,)$ to obtain a similarity score between the prototype $P_{m,n}$ and the query feature $f_\phi(\tilde{x}) \in R^{C*H*W}$ as in Equation (3).

$$s_{m,n} = \frac{\exp\left(-d\left(f_\phi(\tilde{x}), P_{m,n}\right)\right)}{\sum_{n'} \exp\left(-d\left(f_\phi(\tilde{x}), P_{m,n'}\right)\right)} \tag{3}$$

where the $s_{m,n}$ represents the similarity score between the query feature $f_\phi(\tilde{x})$ and the nth prototype in the m-group.

During the training stage, we aggregate all scores to one final predictive probability distribution $s' \in R^N$ by average weighting as in Equation (4).

$$s'_n = \frac{1}{M} \sum_{i=m}^{M} s_{m,n} \tag{4}$$

The classification loss function can be defined as follows:

$$\mathcal{L}_c(\tilde{x}, \tilde{y}; \phi, w_c) = -\frac{1}{N * B} \sum_{j=1}^{N*B} \sum_{n=1}^{N} \tilde{y}_{j,n} \log_{w_c}\left(s'_{j,n}\right) \tag{5}$$

where $(\tilde{x}, \tilde{y}) \in Q$, $\tilde{y}_{j,n}$ represents the ground truth of $\tilde{x}_j$, and $s'_{j,n}$ is the predictive probability of $\tilde{x}_j$ for P_n. w_c is the weight of the loss function, and the Equation (5) represents the cross entropy loss function.

The self-supervised classifier module allows the embedding function to learn the amount of transformations applied in input samples. Specifically, our self-supervised module improves the adaptive capacity of the primary classification module by using a M linear classifier $\sigma : \mathbb{R}^{C*H*W} \to \mathbb{R}^M$. The self-supervised loss function is defined as follows:

$$\mathcal{L}_t\left(x, y'_{ssl}; \phi, w_t\right) = \mathcal{L}_{CE}\left(\sigma\left(f_\phi(x), w_t\right), y'_{ssl}\right) \tag{6}$$

where y'_{ssl} represents self-supervision labels which refers to the different transformations. And w_t is the parameter of the linear classifier and $\mathcal{L}_{CE}$ is another cross entropy loss function.

Finally, we combine the classification loss and the self-supervised loss to learn our final collaborative loss with a hyper-parameter β.

$$\mathcal{L}_{MT}(\phi, w_c, w_t) = \mathcal{L}_c(\phi, w_c) + \beta * \mathcal{L}_t(\phi, w_t) \tag{7}$$

where β is an empirical value, which will be discussed in Section 4.4. The collaborative loss regularises our model in a various amount of image transformations and allows the model to learn a robust classifier with limitedly available samples.

3.4 | Model-agnostic ensemble inference for decision fusion

In this subsection, we discuss our MAEI mechanism for FSL. The results of FSL methods have low confidence scores due to limited samples. Therefore, we consider using MAEI mechanism to eliminate the intrinsical uncertainty for FSL. The main idea of our MAEI mechanism is suppressing the non-maximum similarity scores and only preserving the maximum one.

NMS [39] is proposed to remain the boxes with high confidence as detection results for the similar point, which can benefit to compute intersection over union (IoU) for the

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object detection problem [40]. Analogy to our classification task, we think each predictive label can be seen as a box, and the low confidence predictive labels may cause the uncertainty of our model. Therefore, we attempt to perform NMS to remove the low confidence predictive labels and obtain a reliable ensemble result. Specifically, during the testing stage, we suppress the non-maximum score in the same group, then the preserved scores could be aggregated by average weighting. The NMS can be formalised as follows:

$$s_{m,n} = \begin{cases} s_{m,n} & \text{if } s_{m,n} \text{ is maximum in the m-th group} \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

MAEI is a simple, flexible, and effective mechanism for FSL in the image classification. And we show the superiority of the MAEI mechanism by an example: as shown in the Figure 4a, there are four groups similarity scores, and the first three groups are predicted correctly (the blue score is the highest), but the final result predicts unsuccessfully because the 4-th group similarity score is an outlier compared with others. As shown in Figure 4b, our method has a correct result, since our MAEI mechanism can eliminate the influence of outlier by the NMS. Additionally, our MAEI mechanism can also be applied in other metric-based methods, which will be discussed in Figure 5. And the details of the testing are described in Algorithm 1.

Algorithm 1 MAEI for the FSL tasks during the testing stage

Require: f_ϕ: The embedding function with learned parameters
Require: $H()$: The composed augmentation operation
Load the learned parameters ϕ
Sample the support and query set $\{S, Q\}$ to construct tasks T from the test/val data set
for all $t \in T$ **do**
  Load S, $Q \in t$
  Obtain S_{aug} by Equation (1)
  Divide S_{aug} to M groups and define S_m as the m-th group, $m \in \{1, 2, \ldots, M\}$
  Get the support embedding features from per group: $f_\phi(x_{i,j})$, $x_{i,j} \in S_m$
  Get the query embedding feature: $f_\phi(\tilde{x})$, $\tilde{x} \in Q$
  Obtain the prototype $p_{m,n}$ by Equation (2)
  Compute the similarity score $s_{m,n}$ between the $p_{m,n}$ and the query embedding feature $f_\phi(\tilde{x})$ by Equation (3)
  Perform the NMS operation for each $s_{m,n}$ by Equation (8)
  Compute the final predictive label s' by Equation (4)
  Compute the accuracy for this test/val
end for

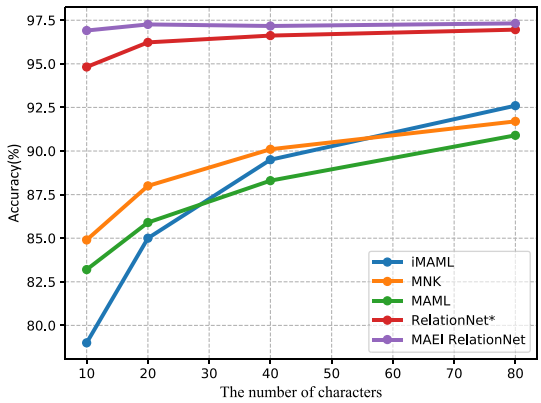
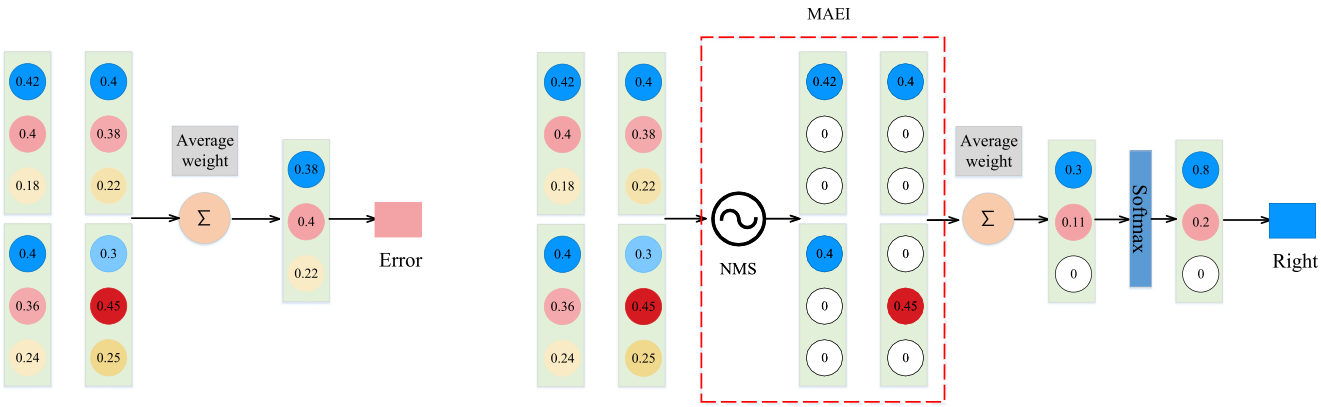


FIGURE 5 The 5-*way* 1-*shot* results on Omniglot data set with the MAEI mechanism on RelationNet. '*' represents results from retraining. MAEI, *model-agnostic ensemble inference*.



(a) An unsuccessful prediction without MAEI **(b)** A successful prediction with MAEI

FIGURE 4 The difference prediction mechanism with or without MAEI. MAEI, *model-agnostic ensemble inference*.

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4 | EXPERIMENTS

In this section, we conduct some few-shot experiments to validate the performance of our method. We first introduce the detail of data sets and show the main experimental settings. Then, we present several comparative results with other state-of-the-art methods to demonstrate that our method can achieve the most excellent performance. Finally, we perform ablation analyses to better understanding the performance of each component of our method.

4.1 | Data sets and metrics

We validate our method on Mini-ImageNet [22], Tiered-ImageNet [41], FC100 [42], and Omniglot data set [1] under a common experiment metric. Specifically, we perform the common *N-way K-shot* task for example, 5-*way* 1-*shot* and 5-*way* 5-*shot*, on Mini-ImageNet, Tiered-ImageNet, and FC100 data sets. We do some ablation analyses on Mini-ImageNet and Omniglot data sets.

Mini-ImageNet: Mini-ImageNet, originally proposed by Vinyals [22], is a common data set to validate different few-shot classification approaches. It has high complexity since this data set is randomly selected from the ImageNet [43], which is a large scale data set for image recognition applications. Mini-ImageNet data set contains 100 categories of objects with 600 samples per category. Following a commonly-used split strategy, we resize each image to 84 × 84 pixels and use 64/16/20 classes for train/val/test respectively.

Tiered-ImageNet: Tiered-ImageNet is an another subset of ImageNet [43]. Different from Mini-ImageNet, Tiered-ImageNet is a larger scale standard benchmark for FSL tasks and generates the splits of disjoint classes based on WordNet. In total, it contains 608 different classes and further splits into 34 super-classes. Following the split method of SLA [7], we use 20/6/8 super-classes (i.e. 351/97/160 low-level classes) for train/val/test respectively. In addition, all images are resized to 84 × 84 pixels before being fed into our model.

FC100: Based on CIFAR100, the FC100 data set is proposed by Oreshkin et al. [42] for FSL first. FC100 reduces information overlap by reorganising 100 different classes from the CIFAR100 data set to 20 super-classes. Therefore, it offers a more challenging problem in few-shot recognition applications due to smaller resolution and stricter class split than Mini-ImageNet data set. Following the split strategy of Lifchitz et al. [5], all images are resized to 32 × 32 pixels and use 12/4/4 classes for train/val/test.

Omniglot: The Omniglot data set [1] is previously proposed to develop the one-shot learning task, which contains 1623 various handwritten symbols compiled by 20 different people. It is a simple benchmark for the few-shot image classification task, which possesses a similar complexity with the MNIST data set. The normal split strategy uses 1200 characters for training and the rest for testing. However, just like Wang et al. [44], we use a small training data set to simulate human learning. Specifically, we perform the 5-*way* 1-*shot* classification task with 10, 20, 40, 80 characters and the rest for testing. Similarly, all images are resized to 28 × 28 pixels following [10].

Implementation Details: Firstly, we perform our experiments using ResNet-12 as the encoding function, and our model uses end-to-end training strategy. Note that, we chose ResNet-12 as a simple backbone to emphasise the method, not the feature extractor. Secondly, the network is optimised via SGD and we train our model for 60,000 episodes in total. For the Omniglot data set, the learning rate is always 0.001 [10]. For the other data sets, the initial learning rate is 0.1, and then changed to 0.001 and 0.0001 at episodes 20,000 and 40,000 respectively. In addition, following the *K-shot* setting of [10], the 5-*way* 1-*shot* task has 15 query images, and the 5-*way* 5-*shot* task contains 10 query images for Mini-ImageNet and Tiered-ImageNet data sets. For FC100 and Omniglot data sets, we set 19 and 15 query images for 5-*way* 1-*shot* and 5-*way* 5-*shot* respectively. The hyper-parameter β is 0.4 and M is 4, 4, 4, 6 for mixup/rotation, mixup/colour permutation, rotation/colour permutation, and mixup/rotation/colour permutation respectively. Finally, all experiments are performed on PyTorch with two 11 GB 2080Ti GPU.

4.2 | Main results

Results for FSL: To evaluate the performance of our model, we present a comparison of the proposed approach with 12 other existing state-of-the-art methods on three benchmark data sets for the 5-*way* 1-*shot* task and the 5-*way* 5-*shot* task.

As shown in Table 2, our method could outperform prior methods and achieve the best results the 5-*way* 5-*shot* task on the Mini-ImageNet data set. Specifically, our method is approximately 0.71% higher than the second-best approach on Mini-ImageNet data set for the 5-*way* 5-*shot* task. In addition, our method can also achieve the best results the 5-*way* 1-*shot* task on the Teried-ImageNet data set, which is 0.24% higher than the second-best approach. Similarly, we also achieve state-of-the-art on the FC100 data set for the 5-*way* 1-*shot* task and the 5-*way* 5-*shot* task. Therefore, extensive results demonstrate that our method outperforms prior approaches. Note that, both our method and SLA [7] used aggregated inference strategy during the testing stage, but our results is better than SLA [7].

We believe that our method has three advantages. Firstly, the self-supervised task makes our model learn the variant property of transformation differences, which can improve the performance of primary classification tasks. Secondly, our method is able to effortlessly adapt to the FSL classification task because we explore the best composed augmentation with self-supervision signals. Finally, the MAEI mechanism can further improve the reliability of multi-task learning framework, which is a simple, flexible, and effective mechanism for FSL.

Results for Augmentation Strategy with Self-supervision Signals: Based on the feasibility of multi-task

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TABLE 2 Results for different methods on Mini-ImageNet, Teried-ImageNet, and FC100 data sets

Methods	Backbone	Mini-ImageNet		Teried-ImageNet		FC100	
		1-*shot* (%)	**5-*shot* (%)**	**1-*shot* (%)**	**5-*shot* (%)**	**1-*shot* (%)**	**5-*shot* (%)**
MAML [19]	Conv-64	48.70 ± 1.84	63.11 ± 0.92	58.90 ± 1.9	71.50 ± 1.0	-	-
R2D2 [45]	Conv-64	-	-	65.30 ± 0.2	79.40 ± 0.10	-	-
RelationNet [10]	Conv-64	50.44 ± 0.82	65.32 ± 0.70	55.00 ± 1.6	69.30 ± 0.80	-	-
SNAIL [46]	ResNet-12	55.71 ± 0.99	68.88 ± 0.92	-	-	-	-
TADAM [42]	ResNet-12	58.50 ± 0.30	76.70 ± 0.30	-	-	40.10 ± 0.40	56.10 ± 0.40
Prototypical [23]	Conv-64	50.44 ± 0.82	65.32 ± 0.70	54.50 ± 1.00	68.30 ± 0.86	-	-
Prototypical* [23]	ResNet-12	58.76 ± 0.74	74.32 ± 0.28	72.36 ± 0.68	83.71 ± 0.47	37.34 ± 0.42	52.62 ± 0.54
MetaOptNet [21]	ResNet-12	62.64 ± 0.61	78.63 ± 0.46	72.00 ± 0.70	84.20 ± 0.50	41.10 ± 0.60	55.50 ± 0.60
SLA [7]	ResNet-12	62.22 ± 0.69	77.78 ± 0.51	74.60 ± 0.70	86.80 ± 0.50	40.00 ± 0.60	55.70 ± 0.60
S2M2R [11]	WRN-28-10	64.93 ± 0.18	83.18 ± 0.11	73.71 ± 0.22	**88.59 ± 0.14**	-	-
RENet [14]	ResNet-12	**67.60 ± 0.44**	82.58 ± 0.30	71.61 ± 0.51	85.28 ± 0.35	-	-
CC+rot [29]	WRN-28-10	62.93 ± 0.45	79.87 ± 0.33	70.53 ± 0.51	84.98 ± 0.36	-	-
SKD-GEN1 [30]	ResNet-12	67.04 ± 0.85	83.54 ± 0.54	72.03 ± 0.91	86.50 ± 0.58	-	-
Ours	ResNet-12	66.72 ± 0.36	**84.25 ± 0.46**	**74.84 ± 0.18**	87.52 ± 0.43	**41.72 ± 0.32**	**56.32 ± 0.52**

Note: Each experiment is tested over 1000 episodes and averaged to get the results. The best results are highlighted by bold.

learning, we perform several experiments to explore the best composed augmentation strategy with self-supervision signals for FSL on Mini-ImageNet. We use the Prototypical network [23] without the self-supervised learning strategy as the baseline. Then, we attempt to predict three data augmentation operations as self-supervision signals such as rotation, colour permutation, and mixup, and explore the best combination augmentation operation for the few-shot image classification. As shown in Table 3, we divide all experiments into two groups: single data augmentation and composed data augmentations.

In the single data augmentation, we verify different data augmentation strategies during the training and testing stage to discuss which combination strategy is more suitable for the FSL task. In general, three different operations are used in training and testing phase, and there are $3 \times 3 = 9$ possible combinations in total. As shown in the middle rows of Table 3, we find that using the same operation in training and testing is able to achieve the best results for each data augmentation strategy. Specifically, the best results are 63.46% and 80.63%, 62.29% and 78.43%, 61.34% and 76.62% for the mixup, rotation, and colour permutation respectively.

In the composed data augmentations, we perform different operations for input images both in training and testing. We find that the combination of mixup and rotation operation with self-supervision signals can achieve the best results, which are 66.72% and 84.25% for the 5-*way* 1-*shot* task and the 5-*way* 5-*shot* task respectively.

In a nutshell, all data augmentation operations are able to achieve positive results for the FSL task, which can further verify the effectiveness of our method. We also observe that the same augmentation operation both in training and testing can better adapt to the FSL classification task. The reason may be that the supervision signal from self-supervised learning can be more effective under the same operation in training and testing stages.

TABLE 3 Different data augmentation results on Mini-ImageNet data set

	Train			Test				
	Mix	**Rot**	**Col**	**Mix**	**Rot**	**Col**	**1-*shot* (%)**	**5-*shot* (%)**
Baseline							58.76 ± 0.74	74.32 ± 0.28
Single			√			√	61.34 ± 0.23	76.62 ± 0.49
			√		√		60.28 ± 0.86	75.23 ± 0.78
			√	√			60.92 ± 0.66	76.32 ± 0.14
		√				√	61.68 ± 0.34	76.35 ± 0.37
		√			√		62.29 ± 0.56	78.43 ± 0.47
		√		√			61.90 ± 0.62	77.06 ± 0.24
	√					√	61.68 ± 0.64	76.03 ± 0.48
	√				√		62.05 ± 0.67	77.52 ± 0.50
	√			√			63.46 ± 0.56	80.63 ± 0.33
Composed		√	√		√	√	64.31 ± 0.59	81.76 ± 0.55
	√		√	√		√	65.58 ± 0.58	83.04 ± 0.48
	√	√	√	√	√	√	65.47 ± 0.48	82.94 ± 0.34
	√	√		√	√		**66.72 ± 0.36**	**84.25 ± 0.46**

Note: 'Mix', 'Rot', 'Col' represent mixup, rotation, and colour permutation operation respectively. The best results are highlight by bold.

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4.3 | Ablation experiments

In this subsection, we perform some ablation experiments comparing with Prototypical network [23] (58.76% and 74.32% for 5-*way* 1-*shot* and 5-*way* 5-*shot* respectively) to demonstrate the effectiveness of the composed augmentation, the multi-task for FSL, and the MAEI on Mini-ImageNet data set for 5-*way* *K*-*shot* tasks. To further verify the adaptive capacity of the MAEI module, we also transfer it to RelationNet [10] on Omniglot data set for the 5-*way* 1-*shot* task.

As shown in Table 4, the 'CA', 'MTFSL', and 'MAEI' represent the composed augmentation, multi-task for FSL, and the MAEI respectively. Our baseline is Prototypical network [23] based on ResNet-12 without these three strategies. The Prototypical network [23] with CA has a little improvement compared with the baseline. In addition, the MTFSL can get prominent results due to the capability representation of the augmented images. And adding the MAEI can further improve the accuracy by introducing a more reliable result.

To further evaluate the generalisation performance of MAEI, we transfer the MAEI module to RelationNet and test on the Omniglot data set. We name the new method as MAEI RelationNet, and compare it to other existing four approaches: MAML [19], iMAML [47], MNK [44], and RelationNet [10]. All four approaches belong to meta-learning methods, MAML aims to find an optimal gradient descent point as pre-trained model to fast adapt to novel tasks. iMAML and MNK are the variant of MAML. As shown in Figure 5, we can obverse that the MAEL RelationNet can achieve the best results based on the same experiment setting. Therefore, the MAEI mechanism possesses great generalisation performance.

TABLE 4 Ablation experiment results of our method

CA	MTFSL	MAEI	1-*shot* (%)	5-*shot* (%)
			58.76 ± 0.74	74.32 ± 0.28
√			59.26 ± 0.27	75.09 ± 0.37
√	√		63.58 ± 0.34	80.19 ± 0.56
√	√	√	66.72 ± 0.36	84.25 ± 0.46

Note: Our baseline is Prototypical network [23] without these three strategies.

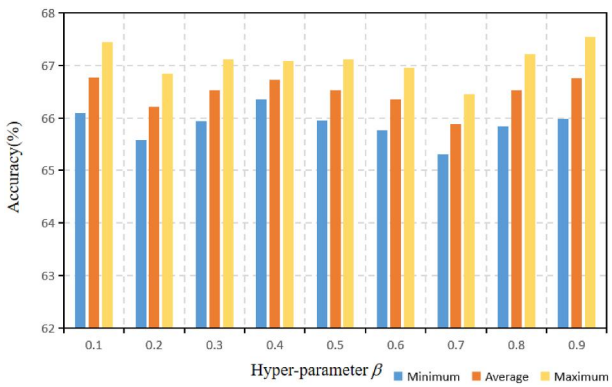

FIGURE 6 5-*way* 1-*shot* results on the Mini-ImageNet data set with different hyper-parameter β.

4.4 | Hyper-parameter analysis

The hyper-parameter β controls the ratio of the importance between the classification loss and the self-supervised loss. We evaluated β on the 5-*way* 1-*shot* task on the Mini-ImageNet data set. As shown in Figure 6, we test nine different values of β from 0.1 to 0.9, and exhibit these average accuracy and corresponding variance from over 1000 episodes. We can observe that our model is stable in different choice of β (all average results are over 66% expect $\beta = 0.7$ [65.88 ± 0.58%]). Specifically, when $\beta = 0.1$ and $\beta = 0.4$, our method achieves the best and second best results. Based on the above results, we select $\beta = 0.4$ as the final result because its variance is smaller and the average result is the second best result.

5 | CONCLUSIONS

In this paper, we introduce a multi-task learning framework for FSL, which consists of a primary classification task and an auxiliary self-supervised task. Our multi-task learning framework can learn the transformation property as auxiliary self-supervision signals to improve the performance of the primary few-shot classification task. In addition, we use three data augmentation operations as self-supervision signals to construct the best combination augmentation operations for few-shot image classification. In order to further improve the reliability of the classifier, we propose a MAEI mechanism for decision fusion. This mechanism is a plug-and-play technology, which can adapt to any metric-based FSL methods without fine-tuning. Our extensive analysis and experiments show that the MAEI mechanism can make a significant performance improvement to existing FSL methods. We believe that our work can bring in many interesting directions for future researches. For example, we can further explore the interplay between the metric learning elements of FSL with data augmentation, which can also generalise to the unsupervised task. As future work, we also see great potential on analysing the effect of larger backbones.

AUTHOR CONTRIBUTIONS

Rui Zhang: Conceptualisation; Methodology; Supervision; Validation. **Yixin Yang:** Conceptualisation; Methodology; Writing – Original draft preparation. **Yang Li:** Conceptualisation; Methodology; Project administration; Writing – review & editing. **Jiabao Wang:** Validation; Investigation; Software. **Hang Li:** Writing – Original draft preparation; Validation. **Zhuang Miao:** Writing – review & editing; Supervision.

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CONFLICT OF INTEREST

The author declares that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed in this study.

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